

Complete Literature Review

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

1. Introduction

Predictive maintenance (PdM) applies condition-monitoring, operational and sensor data to predict equipment failures, identify degradation and estimate remaining useful life (RUL) before machinery reaches an unacceptable operating condition. Therefore, significant research into data-driven fault diagnosis, machine-failure classification, and prognostics has been stimulated by the growing availability of Industrial Internet of Things devices, linked sensors, and machine-learning (ML) technology. According to systematic evaluations, current PdM research encompasses traditional machine learning (ML), deep learning (DL), hybrid techniques, prognostics, condition monitoring, and more complicated Industry 4.0 deployment scenarios [1]-[4].

Traditionally, much of the present research has focused on developing or comparing predictive models. Tree-based ensemble models such as Random Forest (RF) and gradient-boosted methods such as XGBoost are widely applied to structured industrial data, whereas recurrent architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks are particularly relevant to sequential degradation and RUL estimation. These comparisons are required because an effective maintenance system still requires sufficient predictive capabilities.

However, when the goal switches from benchmark prediction to deployment-oriented model selection, the research challenge becomes different. The best configuration for deployment may not always be the model with the greatest F1-score, area under the receiver operating characteristic curve, or lowest RMSE. Reliability of projected probabilities, inference efficiency, robustness to degraded sensor inputs, and the operational effects of prediction mistakes are other factors that may influence operational decisions. Real-world robustness, dynamic operating conditions, diverse data, and implementation requirements are still identified in recent PdM research as the main obstacles separating promising benchmark performance from real-world deployment [1], [5].

Therefore, the literature reviewed throughout this research project supports a revised research narrative. The central problem is no longer simply: Which predictive-maintenance algorithm produces the strongest benchmark performance?

Instead, the problem becomes: How should several potentially conflicting sources of deployment evidence be combined to produce a defensible model recommendation?

Besides, further complication arises because the empirical evidence supporting different deployment criteria may not be equally reproducible. A ranking based on one criterion may remain highly consistent across repeated evaluations, whereas another criterion may produce a substantially less stable ordering of the same candidate configurations. Hence, treating both rankings as equally trustworthy can discard information concerning the reliability of the underlying evidence.

Existing research already provides methods for predictive evaluation, robustness testing, probability calibration, multi-criteria decision making (MCDM), rank aggregation and uncertainty analysis. Consequently, the research gap is not that these techniques are absent. Rather, the gap lies at their intersection: how empirical differences in the reliability of criterion-level rankings should influence deployment-oriented rank aggregation.

This review develops the research direction after synthesising the literature in five related topics. Predictive-maintenance modelling and benchmark evaluation, robustness, calibration, and deployment limitations, MCDM and rank aggregation, uncertainty and ranking reliability, and the literature basis for Reliability-Aware Deployment Rank Aggregation (RDRA) are all examined. The resulting synthesis is used to define the research gap and justify the intended research direction.

2. Predictive-Maintenance Modelling and Benchmark Evaluation

2.1 Data-driven predictive-maintenance modelling

Numerous statistical, machine learning, and deep learning techniques have been used in PdM research. Reviews by Zonta et al. [2], Zhang et al. [3], and Lei et al. [4] show the variety of methods utilised for fault prediction, RUL estimation, and equipment-condition evaluation.

Traditional supervised-learning approaches such as logistic regression, support vector machines, decision trees and random forest remain relevant because industrial datasets are frequently structured, heterogeneous and moderate in size. Ensemble-tree methods are especially useful where nonlinear relationships and interactions among sensor measurements influence failure behaviour.

Gradient-boosting approaches such as XGBoost have similarly become crucial because they combine strong predictive performance with efficient optimisation and the ability to model complex nonlinear feature relationships. In contrast, sequential degradation data motivates architectures capable of learning temporal dependencies. LSTM and GRU networks have therefore been widely investigated for RUL prediction and other temporal prognostics problems.

Crucially, the literature does not identify one universally dominant PdM algorithm. Model suitability depends on the prediction task, data characteristics, evaluation objective and operating environment [1]-[5]. This indicates that modern PdM research increasingly generates

not only a prediction problem, but also a model-selection problem involving multiple technically credible alternatives.

This observation directly relates to the present research. The goal is not to develop another failure classifier or RUL model. Instead, several candidate algorithms and configurations can be considered as alternatives whose suitability must be evaluated across broader deployment requirements.

2.2 Limitation of performance-only comparison

Conventional model evaluation normally emphasises predictive performance. Classification studies frequently report accuracy, precision, recall, F1-score and area-under-curve metrics, while RUL studies commonly use RMSE and MAE.

These metrics remain essential. A model that does not provide sufficient predictive capability is unlikely to become an appropriate deployment candidate merely because it performs well on secondary dimensions. However, deployment suitability cannot be adequately represented by prediction performance alone.

For instance, two classifiers can exhibit similar F1-scores while differing significantly in their behaviour under noisy sensor inputs. Similarly, a model may achieve strong discrimination while producing poorly calibrated probabilities. Another configuration may provide slightly lower predictive performance but substantially lower inference cost.

Thus, a benchmark score should be interpreted as one source of evidence rather than as a complete model-selection rule.

This leads to an important distinction:

$$\textit{prediction performance} \neq \textit{deployment suitability}$$

Recent research increasingly acknowledges this distinction through stronger emphasis on robustness, implementation constraints and real-world validation [1], [5].

2.3 AI4I and C-MAPSS as complementary benchmark evidence

Benchmark datasets are important because they allow controlled and reproducible comparison. Therefore, two datasets are particularly relevant to the present research.

The AI4I 2020 Predictive Maintenance Dataset, which comprises 10,000 observations with variables such as air temperature, process temperature, rotational speed, torque, and tool wear, was created as a synthetic representation of industrial machine-condition data [6]. It supports machine-failure classification and provides a convenient environment for controlled experimentation.

AI4I is helpful since it allows for the recurrent evaluation of several candidate configurations under similar circumstances. Systematic introduction of controlled data-quality corruption is another option. However, promises of universal industrial generalization are limited by its synthetic nature.

The NASA C-MAPSS turbofan-engine benchmark addresses a different problem. It is frequently used for RUL estimation and includes simulated engine degradation trajectories. C-MAPSS is a sequential prognostics problem, while AI4I represents structured machine-failure classification.

As a result, these datasets provide complementary rather than identical evidence. Cross-task evaluation is beneficial because it reduces dependence on a single prediction problem and helps determine whether deployment-oriented conclusions are specific to one benchmark setting.

Synthesis and remaining limitation

Strong predictive modeling techniques, benchmark datasets, and traditional evaluation measures have all been sufficiently established in the PdM literature. However, the growing availability of multiple capable models means that model selection increasingly depends on requirements beyond predictive accuracy.

Furthermore, any model ranking derived from a benchmark is still dependent on the metric, evaluation process, and dataset selected. Therefore, rather than being an indisputable ground truth, a criterion ranking should be understood as empirical data generated by an evaluation process. This becomes crucial when many criterion-specific rankings need to be aggregated.

3. Robustness, Calibration and Deployment Constraints

3.1 Robustness under sensor and data-quality degradation

PdM systems deployed in operational environments depend on physical sensors and data-acquisition infrastructure. Unlike clean benchmark datasets, operational data can contain measurement noise, unavailable observations, abnormal spikes, systematic offsets, drift and other forms of degradation.

Recent PdM research continues to identify data quality and robustness as major deployment challenges [1]. Fault-diagnosis research under noisy conditions similarly demonstrates that predictive models can behave differently when signal quality deteriorates [7].

Consequently, robustness evaluation should not be limited to a single clean test set. Controlled perturbation experiments provide a more informative mechanism by introducing specific forms of sensor degradation at increasing severity levels.

Examples include:

- Gaussian noise
- missing observations
- outliers
- static sensor offsets
- drift
- frozen measurements

Thus, robustness should be viewed as conditional on an evaluation protocol:

$$R = f(\text{corruption type}, \text{severity}, \text{metric}, \text{model})$$

This means that robustness itself is not necessarily represented adequately by one universal scalar score. A candidate may perform well with random noise but substantially worse under missing values. Another candidate may exhibit the reverse pattern. Conclusions can also vary with corruption severity.

This creates a second-level problem for model selection. The important question is no longer only: How much predictive performance is lost after corruption?

It is also: How consistently does robustness evidence rank the competing model configurations?

The criterion ranking is supported by less reproducible evidence if the resulting ordering varies significantly under reasonable experimental settings. This distinction becomes central to the present study because RDRA is concerned with the reliability of criterion-ranking evidence, not merely individual model robustness.

Synthesis and remaining limitation

The robustness of literature strongly supports evaluating model behaviour under degraded data. What is less directly addressed in deployment-oriented PdM selection is how the empirical stability of robustness-derived rankings should affect their influence within a final consensus ranking.

3.2 Probability reliability and calibration

Different aspects of model quality are measured by probability calibration and predictive discrimination.

A classifier can correctly rank positive cases above negative cases while still producing probability values that are systematically too high or too low. This distinction is important when maintenance actions or decision thresholds are influenced by predictive probability.

Niculescu-Mizil and Caruana [8] illustrated the utility of Platt scaling and isotonic regression and showed that supervised-learning systems can display typical probability distortions. Strong

predictive performance does not always imply well-calibrated confidence estimations, as Guo et al. [9] subsequently demonstrated. Similarly, Van Calster et al. [10] contends that whenever probability estimates are used to inform decisions, calibration is a crucial part of predictive analytics.

This directly relates to PdM. Whether equipment is inspected, whether maintenance action is planned, or whether an alarm threshold is exceeded can all be influenced by a predicted probability of failure. Therefore, the quality of the predicted probability matters in addition to whether a binary label is correct.

Techniques including the Brier score, reliability curves, and expected calibration error can be used to evaluate calibration. Alternative configurations of the same underlying predictive model can be generated by post-hoc techniques such as isotonic regression and Platt scaling.

Therefore, a model configuration rather than just an algorithm family should be the unit of deployment comparison.

For instance,

XGBoost + no calibration

and

XGBoost + isotonic calibration

are distinct deployment candidates because they may differ in probability reliability and potentially in other operational characteristics.

Synthesis and remaining limitation

Calibration literature adequately establishes why predicted-probability reliability matters and provides mature techniques for evaluating and improving it. The remaining deployment-selection issue is how probability-reliability evidence should be reconciled with predictive quality, corruption robustness, efficiency and operational consequences when the rankings disagree.

3.3 Computational efficiency and operational consequences

Hardware and software limitations may also have an impact on deployment suitability. Candidate configurations may differ in inference latency, memory consumption, model size and computational complexity. These properties can become especially important when predictions are generated frequently, executed close to equipment, or deployed in constrained environments.

A computationally demanding model may achieve marginally better predictive performance while being less suitable for continuous operation. Conversely, a lightweight configuration may sacrifice some predictive performance but provide substantial deployment benefits.

Operational consequences introduce another source of heterogeneity. False-positive and false-negative predictions do not necessarily have equivalent impacts. Failure to detect an emerging fault may lead to equipment failure or downtime, whereas an unnecessary warning can trigger additional inspection or maintenance activity.

Lv et al. [11] explicitly consider predictive-maintenance decision making under non-equivalent fault costs, illustrating that maintenance consequences cannot always be represented adequately through generic classification metrics.

Therefore, deployment-oriented model selection can be conceptualised as:

$$\text{Deployment Suitability} = f(P, R, C, E, O)$$

where:

- P = predictive quality
- R = robustness
- C = probability reliability or calibration
- E = computational efficiency
- O = operational impact

This corresponds closely to the five deployment dimensions used in the current research: predictive quality, robustness, probability reliability, efficiency and operational impact.

Synthesis

Robustness, probability reliability, computational efficiency and operational consequences provide legitimate deployment evidence beyond conventional predictive metrics. However, evaluating more dimensions does not automatically solve the selection problem. It creates a multi-criteria decision problem in which heterogeneous evidence must be combined transparently and defensibly.

4. Multi-Criteria Decision Making and Rank Aggregation

4.1 Multi-criteria decision making

Multi-criteria decision making (MCDM) is a mature research area concerned with decisions involving competing criteria. Liao et al. [12] review data-driven MCDM and its increasing interaction with machine learning, showing that MCDM provides systematic mechanisms for comparing alternatives under heterogeneous criteria. Widely used techniques include:

- Analytic Hierarchy Process
- TOPSIS
- VIKOR

- ELECTRE
- PROMETHEE
- Best-Worst Method

These methods differ in how they represent preference, compromise, outranking or distance from desirable alternatives.

Pareto analysis provides another useful perspective by identifying alternatives that are not dominated simultaneously across all objectives. A candidate is Pareto dominated when another alternative performs at least as well on every objective considered and better on at least one.

Pareto analysis is useful for eliminating clearly inferior alternatives but does not necessarily identify a single best option. Several non-dominated solutions can remain, each representing a different trade-off. This is precisely why a consensus or decision mechanism is required.

4.2 Meaning of criterion weights

Weighting is central to many MCDM approaches, but the meaning of a weight requires careful interpretation.

In conventional MCDM, criterion weights frequently represent importance or preference.

For example, stakeholders might specify:

$$w_{safety} > w_{cost}$$

because safety is considered more important.

However, importance and empirical reliability are conceptually different. A criterion may be operationally important while the available experimental evidence supporting its ranking is comparatively unstable. Conversely, a criterion ranking may be highly reproducible even though a stakeholder considers that criterion less important for a particular deployment.

Therefore,

$$criterion\ importance \neq criterion\ evidence\ reliability$$

Boix-Cots et al. [13] demonstrate that weighting mechanisms across group decision-making systems can arise from different information sources and serve different purposes. Their work reinforces the need to define clearly what weights mean rather than treating weight as a generic operation.

This distinction is fundamental to the present research. RDRA weights are not intended to state Reliability is more important than robustness. Instead, they represent the experimental evidence supporting one criterion ranking is more reproducible than the evidence supporting another.

Stakeholder preference can then be examined separately. This conceptual separation prevents empirical evidence and confidence and subjective importance from being conflated.

4.3 Rank aggregation

The problem can alternatively be expressed as rank aggregation when each deployment condition generates a ranking of candidate configurations.

Rank aggregation creates a single consensus ordering by combining many input rankings.

Rankings have the important benefit of converting disparate criterion measurements into a shared ordinal representation. This avoids directly combining incompatible numerical scales such as latency, calibration error and classification performance.

One well-known rank-aggregation method is the Borda count. A consensus is created by adding the rank-derived scores that each candidates obtains under each input ordering.

Rank-based methods are also applicable to algorithm-selection contexts, where many sources of model-selection information must be balanced, as demonstrated by Tornede et al. [14].

For m criteria, Equal Borda can be expressed as:

$$w_j = \frac{1}{m}$$

This produces a transparent and easily interpretable consensus.

However, Equal Borda makes a consequential assumption: Every criterion ranking receives identical aggregation influence.

This assumption is reasonable where there is no basis for treating one ranking as more reliable than another.

It becomes more questionable when criterion rankings are estimated from empirical experiments and exhibit substantially different reproducibility.

Suppose one criterion ranking is almost unchanged across repeated evaluations while another varies substantially. Equal Borda nevertheless treats them identically.

In this setting, equal aggregation can discard potentially informative evidence concerning ranking reliability.

4.4 Weighted and robust aggregation

Broader literature already contains more sophisticated alternatives to equal aggregation.

An ensemble-ranking method is suggested by Mohammadi and Rezaei [15] for aggregating rankings produced by various MCDM techniques. Their method incorporates metrics of trust and consensus and gives weights objectively. This demonstrates that treating all rankings as equally influential is not always necessary.

However, the meaning of their weight differs from the mechanism considered in this research. Their approach evaluates the degree of agreement between rankings produced by different MCDM techniques. It does not specifically estimate the reproducibility of predictive-maintenance deployment-criterion rankings using repeated experimental evidence.

Kolde et al. [16] propose Robust Rank Aggregation, which evaluates whether items consistently appear near the top of ranked lists more often than expected under a null model. Their approach was developed for gene-list integration and meta-analysis.

Although it shares the phrase “rank aggregation,” its statistical objective differs from RDRA. Robust Rank Aggregation does not directly estimate criterion-specific reproducibility across repeated PdM evaluation evidence and transform that reproducibility into criterion aggregation weights.

These examples are important because they establish an appropriate novelty boundary:

Weighted ranking, robust aggregation and objective ranking weights already exist.

Therefore, the contribution of the present research should not be described as the invention of weighted or reliability-aware ranking in general.

Synthesis and remaining limitation

MCDM and rank-aggregation literature provide mature approaches for combining conflicting information.

However, two issues remain relevant to the present research:

1. Preference-based weighting typically represents criterion importance rather than empirical evidence reliability
2. Equal Borda provides no mechanism for distinguishing criterion rankings according to their reproducibility.

This leads to the need to investigate whether criterion-ranking stability estimated from repeated experimental evidence can provide meaningful reliability information during deployment-oriented aggregation.

5. Evaluation Uncertainty and Ranking Reliability

Machine-learning performance estimates are themselves uncertain.

Variation can arise from:

- sampling
- train-test splits
- random seeds
- model initialisation
- stochastic optimisation
- preprocessing
- data perturbation
- evaluation protocol

Tyralis and Papacharalampous [17] review predictive uncertainty estimation in ML and emphasise that model evaluation should recognise uncertainty rather than relying exclusively on single deterministic estimates.

A similar issue arises when evaluation outcomes are converted to rankings.

Consider a criterion that produces candidate ordering: $A > B > C > D$

If repeated evaluations produce: $A > B > C > D$, $A > C > B > D$ and $B > A > C > D$, the criterion does not provide a perfectly stable ordering.

By contrast, if another criterion produces almost the same ordering across repetitions, its ranking evidence is more reproducible.

Established ranking statistics make this property measurable.

Kendall rank correlation

Kendall correlation evaluates pairwise ordering agreement. It is useful for determining whether pairs of alternatives maintain their relative positions across rankings.

Spearman rank correlation

Spearman correlation assesses monotonic agreement between complete rankings and is useful for evaluating broader rank-order similarity.

Top- k overlap

Top- k overlap measures preservation of the highest-ranked alternatives and is especially relevant where deployment decisions focus on shortlists rather than the complete ranking.

Rank displacement

Mean absolute rank displacement measures the average magnitude by which candidate positions change. These metrics are well-established statistical tools and are not methodological contributions of RDRA.

Their role is instead to provide empirical evidence regarding:

How reproducible is a criterion ranking?

Suppose repeated evaluations of criterion j produce:

$$R_j^{(1)}, R_j^{(2)}, \dots, R_j^{(T)}$$

The degree of agreement among these rankings can be used to estimate a stability quantity: S_j . A larger S_j indicates that the criterion tends to produce a more reproducible ordering of candidate configurations.

This leads to three conceptually different quantities:

- Performance: How well does a candidate perform?
- Preference: How important is the criterion to a stakeholder?
- Evidence reliability: How reproducibly does the evidence rank candidates?

A key insight emerging from literature is that these should not be conflated.

Synthesis and remaining limitation

The broader literature provides established methods for uncertainty estimation and rank-agreement measurement, while decision science provides weighted aggregation approaches.

However, the connection between these areas is less developed in deployment-oriented PdM model selection.

Specifically, there remains a methodological question:

If deployment-criterion rankings are supported by evidence of unequal empirical stability, should those rankings still receive equal influence in the final consensus?

This question directly motivates RDRA.

6. Reliability-Aware Deployment Rank Aggregation (RDRA)

The preceding literature leads to the proposed research direction.

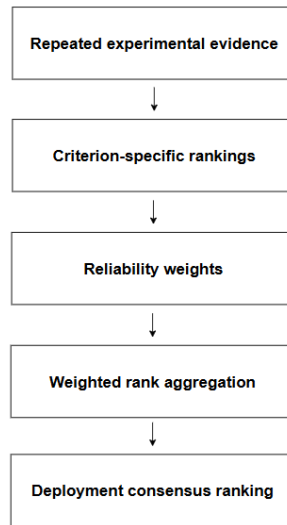
Let S_j represent empirical stability of criterion ranking j .

The reliability information can be normalised as:

$$w_j = \frac{S_j}{\sum_{k=1}^m S_k}$$

Thus, the weight w_j represents relative empirical reliability of ranking evidence. It does not represent subjective stakeholder importance.

Therefore, the conceptual RDRA workflow is:



This formulation is defined in the present research as Reliability-Aware Deployment Rank Aggregation (RDRA).

The novelty boundary needs to remain explicit.

The following components are established:

- Borda scoring
- weighted rank aggregation
- Kendall correlation
- Spearman correlation
- Pareto analysis
- regret analysis
- probability calibration
- robustness testing

The proposed methodological adaptation is instead to estimate criterion-level empirical ranking reliability from repeated predictive-maintenance evaluation evidence and use that reliability as aggregation-weight information in deployment-oriented rank aggregation.

The hypothesised rationale is also specific.

Equal Borda assumes:

$$w_j = \frac{1}{m}$$

RDRA instead allows:

$$w_j \propto s_j$$

Therefore, criterion rankings supported by more reproducible experimental evidence receive relatively greater influence, whereas those supported by less reproducible evidence receive relatively lower influence.

This does not imply that RDRA should be universally superior.

A method that is excessively stable could fail to respond appropriately when meaningful deployment evidence changes. Consequently, good ranking behaviour should not be equated with maximum stability.

A defensible aggregation method should instead be examined from multiple perspectives, including:

- consistency under criterion uncertainty
- responsiveness under data-quality uncertainty
- downstream usefulness under preference uncertainty.

This provides the conceptual bridge from the literature review to the empirical methodology.

7. Literature Synthesis and Research Gap

The main findings from the literature can be consolidated as follows.

Literature area	What existing research addresses well	Remaining issue relevant to this research
Predictive-maintenance modelling	Extensive ML/DL methods for failure prediction and RUL estimation	Predictive performance alone does not establish deployment suitability
Benchmark evaluation	Reproducible datasets and standard predictive metrics	Rankings remain conditional on dataset, metric and protocol
Sensor-data robustness	Controlled corruption and degradation testing	Robustness-derived rankings can themselves vary in stability
Probability calibration	Mature calibration methods and reliability metrics	Calibration evidence must be reconciled with other deployment criteria
Efficiency and operational assessment	Computational and maintenance consequences can be evaluated	These dimensions can conflict with predictive performance
MCDM	Established mechanisms for competing objectives and preference weighting	Criterion importance differs conceptually from evidence reliability
Rank aggregation	Equal, weighted and robust aggregation methods exist	Equal Borda ignores unequal reproducibility of criterion rankings
Uncertainty and ranking agreement	Established uncertainty and rank-correlation measures	Stability information is not automatically incorporated into PdM deployment aggregation
Research opportunity	Relevant technical components exist separately	Investigate whether empirical criterion-ranking reliability can improve deployment-oriented consensus aggregation

Literature therefore does not support a broad claim that deployment evaluation, robustness, calibration or rank aggregation have been ignored.

Research Gap

Therefore, the research gap is existing predictive-maintenance research provides mature methods for predictive modelling, robustness assessment, probability calibration and deployment-oriented evaluation, while MCDM and rank-aggregation literature provides established mechanisms for combining conflicting evidence. However, comparatively less attention has been directed to predictive-maintenance model-selection settings in which the criterion rankings themselves are supported by empirical evidence of unequal reproducibility.

Equal aggregation assigns identical influence on those rankings, while conventional weighting mechanisms primarily represent stakeholder importance or other decision preferences. This creates a methodological opportunity to investigate whether criterion-level empirical ranking stability can be used as evidence-reliability information within deployment-oriented rank aggregation and whether doing so produces a more defensible consensus under evaluation uncertainty.

8. Intended Research Direction

Research Objective

The literature synthesis leads to the following research objective:

To develop and empirically validate a reliability-aware rank-aggregation methodology for deployment-oriented predictive-maintenance model selection that accounts for differences in the stability of evaluation evidence and produces a defensible consensus ranking under evaluation uncertainty.

Research Question

The corresponding main research question is:

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable evaluation evidence?

The proposed methodology, RDRA, evaluates candidate model configurations across five deployment dimensions:

1. predictive quality
2. robustness
3. probability reliability
4. computational efficiency
5. operational impact

Repeated experimental evidence is used to estimate how consistently each dimension ranks the candidate configurations. That stability is then interpreted as evidence-reliability information during consensus aggregation.

Importantly,

RDRA reliability weight $\neq$ stakeholder preference weight

Stakeholder importance is treated separately from empirical reliability.

The empirical research then examines RDRA through three complementary validation perspectives.

VQ1 - Criterion uncertainty

The first perspective evaluates whether RDRA produces a more consistent consensus when the available deployment criteria change.

This addresses the primary intended advantage of reliability-aware aggregation.

VQ2 - Data-quality uncertainty

The second perspective evaluates whether the ranking responds appropriately when the underlying evidence changes because of controlled sensor degradation.

This prevents consistency from being interpreted as rigidity.

VQ3 - Preference uncertainty

The third perspective examines downstream decision usefulness when deployment priorities vary.

This evaluates whether the resulting consensus remains practically informative under different preference conditions.

AI4I provides the primary environment for full RDRA evaluation, including repeated evidence across the deployment dimensions. C-MAPSS provides complementary cross-task evidence from an RUL setting.

Hence, the intended research contribution can be summarised as a deployment-oriented methodological adaptation in which empirical criterion-ranking stability is used as evidence-reliability information during rank aggregation, together with explicit validation of how the resulting consensus behaves under different forms of evaluation uncertainty.

9. Conclusion

The literature review indicates that PdM is an established and fast-growing field. While datasets such as AI4I and NASA C-MAPSS enable reproducible benchmark evaluation, conventional ML and DL techniques offer robust approaches to failure classification, fault diagnosis, and RUL prediction.

Furthermore, the literature shows that deployment suitability goes beyond prediction performance. Additional evidence relevant to real-world model selection is provided by robustness to sensor degradation, probability calibration, computational efficiency, and operational consequences.

This expansion from a single performance metric to multiple deployment criteria creates a new decision problem. Candidate model configurations can obtain conflicting rankings across different dimensions.

MCDM and rank aggregation provide established techniques for combining such evidence. However, the meaning of aggregation weights must be carefully distinguished. Stakeholder preference describes the importance assigned to a criterion, whereas the present research concerns the empirical reliability of the ranking evidence supporting that criterion.

Equal Borda provides a transparent consensus mechanism but gives every criterion ranking equal influence. Where empirical rankings exhibit unequal reproducibility, this equal-influence assumption may discard useful information concerning evidence quality.

Therefore, literature motivates investigation of whether criterion-level empirical ranking stability can serve as reliability information within deployment-oriented rank aggregation.

RDRA operationalises this idea while maintaining a deliberately narrow novelty claim. It does not claim to invent Borda scoring, weighted ranking, rank correlations, calibration or robustness analysis. Rather, it investigates their deployment-oriented integration around the specific problem of unequal criterion-ranking reliability.

The research consequently represents progression from conventional predictive benchmarking toward a more demanding question:

Which model configuration is best supported for deployment when multiple criteria conflict and the empirical evidence supporting those criteria is not equally reliable?

Addressing this question provides a clear pathway from the reviewed literature to the proposed methodology and its subsequent empirical validation.

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